



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid-term Exam, Trimester: Spring 2026

Course Code: CSE 3521 Course Title: Database Management Systems

Total Marks: 30 Duration: 1 Hour 30 Minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. a) You are tasked to build an election management system for the upcoming election. 7
An election commission with a unique id is formed to arrange the elections. Each commission has its own name, email, headquarter address which contains city, street number and zip code information. Each election has a unique election id, with attributes such as election name, date and election type.

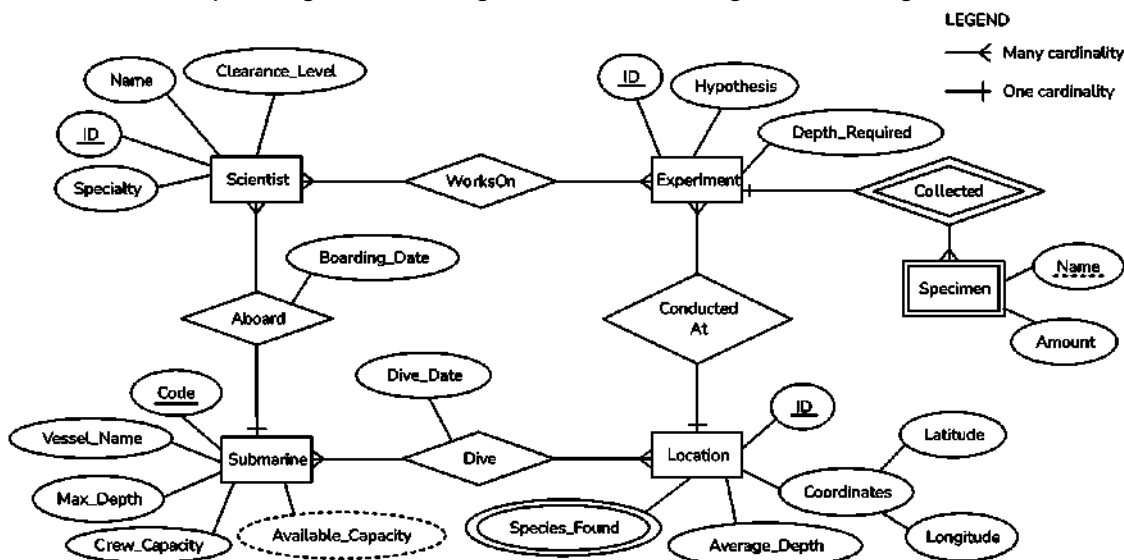
Voters can vote in the elections. Each voter can be identified by NID. Voters have other information such as name, date of birth, age that can be calculated from date of birth, phone number (multiple possible). To identify a particular vote, a vote id is needed along with the nid of the voter and the id of the election. The timestamp of the vote is also saved.

Candidates participate in the elections and are identified by a unique candidate id. Each candidate has his own name, party name, age and symbol. Candidates compete against each other. Each voter is registered at a specific polling station where he can cast vote. Each polling station is uniquely identified by a station id, with attributes such as capacity and location which contains city, street address and zip code. An election commission assigns supervisors at each polling station. The supervisors can be uniquely identified by their id. The age and experience of the supervisor is also tracked.

Now, design an ERD for the system, clearly outlining its entities, attributes and relationships with appropriate cardinality.

- b) Explain the difference between a super key and a candidate key with an example. 2

2. Draw the corresponding schema diagram based on the given ER Diagram. 6



3. Consider the following relational database schema from the famous show 'Tom & Jerry': **1+1+2+1**

Characters(char_id, name, type)

Episodes(episode_id, title, season)

Appearance(char_id (fk), episode_id (fk))

Chases(char_id (fk), episode_id (fk), success_flag)

Answer the following questions using relational algebra:

- a) List the names of all characters whose type is 'Mouse'.
- b) Retrieve the names of characters along with the titles of episodes they appeared in.
- c) Write a sentence describing the output of the following relational algebra:

$\pi_{\text{name}}(\text{Characters}) - \pi_{\text{name}}(\text{Characters} \bowtie_{\text{char_id}} \text{Appearance})$

- d) List the names of characters who have participated in episodes where at least one chase was successful.

4. Use the schema below to answer the following SQL queries:

5x2=10

Player (player_id (PK), name, email, rank, country)

Team (team_id (PK), name, region, created_year)

Tournament (tournament_id (PK), name, game_title, prize_pool, location, start_date)

Match (match_id (PK), tournament_id (FK), team1_id (FK), team2_id (FK), winner_team_id (FK), match_date)

Participation (participation_id (PK), player_id (FK), team_id (FK), join_date)

Sponsor (sponsor_id (PK), name, industry, contribution_amount)

- a) Create the **Tournament** table where name, game_title, and location cannot be NULL.
- b) Add a new column **sponsor_id** to the **Tournament** table and ensure that it references the **Sponsor** table.
- c) Find the **names** and **emails** of **players** who are considered elite and from Japan (the top 20 ranked players are considered elite).
- d) Find the name of the teams that have participated in more than 3 matches.
- e) Retrieve the **tournament name** and **number of matches** played in each tournament, sorted in **descending** order of match count.